The Ribbon at the Bell Bound

Frontmatter

Title: The Ribbon at the Bell Bound: Mapping the Classical Boundary of a Geometric Entanglement Ontology

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Note: Spelling ‘Oktiabrev’ confirmed by author (passport form); contact email spelling differs and is intentional.

Abstract

We report a pre-registered simulation program driving a classical geometric model of entanglement to its quantitative limit. The ontology represents an entangled pair as one framed curve in R^4, its two ends being the observed particles; the framing carries a natural $\mathbb{Z}_2$ topological label, a candidate origin for outcome binarity and spinor structure. The maximal stake — that classical relaxation reproduces singlet statistics — was registered alongside the opposite expectation, in writing, before the first run. The stake was lost, and we map how. A structural theorem shows the $\mathbb{Z}_2$ label is invisible to any axial boundary readout; a census finds no empty-support branch; the amplitude shows no resolved decay over chain lengths 16–96 at two stiffnesses, but is stiffness-controlled and anisotropic — the cosine law survives only along a privileged axis, and honest orientation averaging collapses it to the triangular correlation of Bell’s shared-randomness local model. Two apparent Bell violations ($S = 2.58$, 2.39) died in pre-registered audits: an invalid isotropic estimator, and a setting-dependent post-selection — Pearle’s detection loophole. The corrected family approaches the Bell bound from below ($|S| = 1.62$) and never crosses it. The contribution is the mapped boundary: named, quantified walls between this classical geometry and the singlet state, with a fully version-controlled provenance trail.

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1. Introduction

Take seriously, for the length of one research program, the following picture: an entangled pair is one extended object. Not two particles sharing a state, but a single ribbon embedded in a dimension above the three we see, its two ends being the two intersections with our slice — so that the perfect correlations of the singlet would need no communication, there being nothing to communicate between. The picture has an immediate structural payoff (Section 3): the embedding carries a $\mathbb{Z}_2$ framing class — the framing topology supplies a natural $\mathbb{Z}_2$-valued structural label and the homotopy underlying the 4π belt-trick return. A physical map from this label to measurement outcomes is an additional requirement — and Section 5 proves that for axial readout no such map exists. The bet this program made — its maximal stake — was that the studied classical relaxation model of such an object, read by an honest basin measure, would produce the quantitative statistics of the singlet: the isotropic $-\cos\theta$ correlation at unit amplitude, the Born weights, and a CHSH value of $2\sqrt{2}$. The registered expectation, written into the simulation specification and in the record before the first physics campaign it governs, was the opposite: "the default expectation, stated honestly before launch: a sawtooth or $p \approx 1$; $p = 2$ would be a major result demanding paranoid verification; any outcome is informative …" (SPEC §1, translated from the Russian original; unedited since).[1] The program aimed at the quantum target while predicting, in writing, that it would miss — and it missed exactly where the prediction said it would.

The bet was lost. This paper reports how it was lost — exactly, quantitatively, and, we will argue, usefully. The classical ribbon realizes more of the singlet than we find commonly appreciated: anticorrelation of the right sign, a correlation amplitude that survives chain length unchanged (a kinetic plateau, measured to $N = 96$ and verified at two coupling stiffnesses), and a smooth cosine angular law — though only along one privileged axis. What it cannot realize, we can now name and number: the topology does not enforce exact outcome zeros — the census found no topologically forbidden branch, and the framing invariant is inaccessible to axial readout (a structural theorem: the invariant that could enforce them lives in precisely the fiber that axial measurement quotients out; Section 5); honest isotropy collapses the cosine to the triangular correlation of Bell's shared-randomness local model (Section 6); and every apparent Bell violation the program produced — there were two — died in a pre-registered audit, one to a false symmetry assumption, one to a rediscovery of Pearle's detection loophole (Section 7). The measured family moves toward the Bell bound as stiffness grows ($|S| = 1.62$ at $k\_f\times 4$); its deterministic axial limit is expected — not measured — to reach $\rho \to 1$, $\bar{S} \to 2$, not $2\sqrt{2}$.

We consider the negative answer, so structured, to be the contribution. A classical ontology was driven to a boundary; the boundary was mapped rather than lamented; and the walls were named while the program was still running — several of them named in advance in the pre-registration record. The method that made this possible is plain: hypotheses with kill-criteria committed before runs, raw data committed before analysis, mirror controls, and a standing audit that treats any too-good result as a defect until proven otherwise. That machinery caught not only the physics artifacts above but the authors' own errors, in both directions of the human–AI collaboration that executed the program (Section 9).

[1] Primary program documents (specification, canonical record, campaign reports) are in Russian; quoted excerpts are translated and marked as such.

1.1 Related work and positioning

The walls this program hit are, of course, known walls. Bell's theorem [Bell1964] bounds every local hidden-variable account, and our isotropized correlation is his shared-λ model met dynamically; our causal argument for $\cos\theta$-linearity is, as we found after the fact, the measurement-rule analogue of Gisin's no-go for nonlinear quantum dynamics [Gisin1989; Gisin1990] — there, nonlinearity of the evolution enables superluminal signaling; here, nonlinearity of the outcome rule in $\cos\theta$ does, the argument's form being Gisin's and its object ours — with the uniqueness of the quadratic exponent within the chord family this program's own extension; the post-selection artifact of our second audit is Pearle's detection loophole [Pearle1970]; and the impossibility of eliminating the Born postulate within the symmetric pair is our own methodological observation, standing in the historical line of Kochen–Specker's demonstration that an outcome measure exists [KS1967] rather than following from that theorem. We differ from the classical hidden-variable literature — de Broglie–Bohm's nonlocal dynamics [Bohm1952a; Bohm1952b], 't Hooft's cellular automata [tHooft2016], Palmer's invariant-set theory [Palmer2009] — not in outrunning these theorems, which we do not, but in genre: a single ontology, pursued to its named limit, under pre-registration, with every retreat documented in a version-controlled record — and every rediscovery made after that record began, dated by commit. To our knowledge the combination — an independently conceived geometric ontology, a falsification-first simulation program, and a fully version-controlled provenance trail — has not been reported in this form. The reader who wants the theory is directed to Section 3; the reader who wants the boundary, to Section 6; the reader who suspects that classical models producing $S > 2$ must be hiding an error somewhere will find two such errors, found and dissected, in Section 7.

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2. Model

2.1 Configuration space

The object is a framed discrete curve in R^4: nodes $x_i \in R^4$, $i = 0..N-1$, with unit tangents $t_i = (x_{i+1} - x_i)/|x_{i+1} - x_i|$ and the boundary convention $t_{N-1} := t_{N-2}$. Each node carries a unit quaternion $u_i \in S^3 \subset H$, the lift of the node's normal frame relative to Bishop (minimal-rotation) transport along the curve; the transport $R_i \in SO(4)$ maps t_i to t_{i+1} and acts as identity on the orthogonal complement of their span. The full normal frame — not a single normal vector — is what carries the topology: the normal space of a curve in R^4 is three-dimensional, $\pi_1(SO(3)) = \mathbb{Z}_2$, and the $\mathbb{Z}_2$ framing class lives in the sign of the accumulated lift. A single normal (an S^2 of directions) or a literal two-sided band (an $SO(2)$ fiber) would carry no $\mathbb{Z}_2$ invariant; "ribbon" names the picture, "framed curve" names the object.

2.2 Energy

$H = H_{stretch} + H_{bend} + H_{frame} + H_{clamp}$, with $H_{stretch} = k_s \Sigma (|x_{i+1} - x_i| - \ell)^2$, $H_{bend} = k_b \Sigma (1 - t_i \cdot t_{i+1})$ (harmonic only near alignment), $H_{frame} = k_f \Sigma d^2(u_i, \bar{u}_{i+1})$ where $\bar{u}_{i+1}$ is u_{i+1} parallel- transported back along the Bishop frame and d is the geodesic distance on S^3; the twist datum enters through $\arccos|\langle\cdot,\cdot\rangle|$ and is therefore blind to the lift sign by construction — a modeling choice, not a consequence. Two distinct clamp terms were used and must not be conflated. The frame clamp, $H\_clamp = k\_c \cdot \arccos^2(|\langle u\_end, U\_target\rangle|)$, pins the full boundary frame and appears only in the D0 invariant-validation runs. The axial clamp, $H\_clamp = -k\_c\ (n\_end \cdot a)^2$ with n_end the boundary axis obtained from u_end under $SU(2) \to SO(3)$, pins an axis only — the residual U(1) about it stays free — and is the clamp behind every scientific number in this paper (D1 through the seed audit). The axis–frame distinction is not a technicality: the U(1) theorem of Section 5 lives in exactly the fiber the axial clamp leaves free. The frame energy is blind to the lift sign by construction; the $\mathbb{Z}_2$ invariant is therefore carried by the kinematics — lift continuity between accepted steps and singular-step rejection (2.3–2.4) — not by the energy.

2.3 Dynamics

Relaxation is a projected Euler–Maruyama scheme, stated as implemented: per step, gradients g_x, g_u of H; noise $\nu \sim N(0,1)\cdot\sqrt{(2\cdot lr\cdot T)}$ added in the ambient space; for the frame variables the update $du = -lr\cdot g\_u + \nu$ is projected onto the tangent space at u_i ($du \leftarrow du - \langle du,u\rangle u$) and retracted by normalization. The step size lr doubles as the time step; no metric correction, Itô–Stratonovich term, or retraction Jacobian is applied; consequently we do not claim that the scheme samples a Gibbs measure at temperature T, and no stationary measure is derived — T is an algorithmic noise scale whose observed effect on the reported amplitude is flat (Section 6). Steps are rejected for the whole chain on any of three criteria: a temporal lift jump ($\langle u\_new, u\_old\rangle < 1 - \delta\_sing$), a tangent reversal ($t\_i \cdot t\_{i+1} < -1 + \delta\_tan$), or a spatial lift wall — a sign change of the link datum g.w across one step, the passage of a link through a half-turn ($\delta\_sing = \delta\_tan = 2\cdot 10^{-2}$, band4d.py). The third criterion is what protects the parity bookkeeping: it is the “leaky lift” detector, and the rejection fraction falls with lr in the validated regime. Lift continuity between accepted steps (the sign of u chosen nearest the previous step) is what makes the parity bookkeeping well-defined.

2.4 Preparation and topological invariant

The even sector is prepared as $u \equiv$ const followed by relaxation; the odd sector by a linear twist ramp distributing 2π along the chain, followed by relaxation. The preparation is unbiased with respect to the outcome branch: the two boundary frames are tilted about random axes with polar angle $\arccos(1 - 2\xi)$, so that each end’s axis n_end is uniform on S^2 — the branch is never chosen in advance, and the basin decides (prep_dynamics, measurement.py). Measurement settings a, b are inputs of the protocol, fixed per run or drawn by the campaign design — never from the preparation randomness; their independence from λ is a designed property, used in 2.6(iii). The $\mathbb{Z}_2$ parity of a configuration is the sign of the lift accumulated along the chain with continuous sign choice, compared against the boundary frames; it is defined on trajectories free of rejected singular steps, is exactly conserved on accepted trajectories, and is not spontaneously populated thermally — charged configurations exist only by preparation.

2.5 Measurement and estimator

Outcomes are read locally at the boundaries: $s = \mathrm{sign}(n\_0 \cdot a)$, $t = \mathrm{sign}(n\_{N-1} \cdot b)$, each a function of one end’s frame and that end’s setting only. Configurations with $|n \cdot a|$ below a fixed threshold (0.2, a convention) were classed DEGENERATE and reported as a separate column in the campaigns; for CHSH estimation this class must not be discarded — setting-dependent discard is precisely the detection loophole documented in Section 7, and the corrected estimator keeps all events with sign(0) → +1 by convention. The correlation estimator is $E(a,b) = (1/n\_valid) \Sigma\_\{j: valid\} s\_j t\_j$ with binomial standard error $\sqrt{(\max(1-E^2, \cdot)/n\_valid)}$, where valid excludes the DEGENERATE class where it is reported separately; in the campaigns entering CHSH and the seed audit, degen = 0 and n_valid = M. A pre-registered cross-seed audit measured the seed-to-seed scatter against that binomial error at r = 0.88 (11 repeats, N = 32 core cell) and r = 1.11 (5 points, N = 96), so the quoted errors are honest as reproducibility measures.

2.6 Locality status

Three statements at three strengths, kept separate deliberately. (i) Readout locality holds by construction: s depends on (u_0, a) only, t on (u_{N−1}, b) only. (ii) Dynamical locality is not claimed: both clamps enter one global relaxation functional, so the final configuration near one end may in general depend on the remote setting; proving factorization $P(s,t|a,b,\lambda) = P(s|a,\lambda)\cdot P(t|b,\lambda)$ over the relaxation dynamics would require its stationary measure, which is not derived. (iii) Operationally, the isotropized protocol is a shared-λ scheme: λ = (R, initial configuration and noise realization) with R ~ Haar on SO(3) common to both ends, settings drawn independently of λ, and local readout — and its measured behavior (the triangular correlation, $S \le 2$ everywhere after audit) sits strictly inside the local-hidden-variable region. The paper's Bell-side claims rest on (i) and (iii); nothing rests on (ii).

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3. Analytical layer

3.1 The ontological picture

The ontological picture under test is this: an entangled pair is not two particles bound by a mysterious connection, but a single extended object — a ribbon — embedded in a space of dimension higher than the three we observe. What we call the two particles are the two intersections of this object with our three-dimensional slice; they are boundaries of one body, not separate things. On this picture, the correlations of entanglement would require no communication, because there is nothing to communicate between — one object simply has one geometry. The picture earns its keep at the level of structure: the embedding equips the ribbon with framing classes labeled by $\mathbb{Z}_2$ (since $\pi_1(SO(3)) = \mathbb{Z}_2$). The framing topology supplies a natural $\mathbb{Z}_2$-valued structural label and the homotopy underlying the 4π belt-trick return. A physical map from this label to measurement outcomes is an additional requirement — and Section 5 proves that for axial readout no such map exists.

What this paper tests, to its limit, is whether the classical dynamics of such an object can also produce the quantitative statistics of the singlet state. It cannot — and the ways in which it cannot turn out to be sharply nameable.

3.2 The chord law and the causal derivation of the exponent

The quadratic chord law is a geometric transcription of the quantum target, not a derivation of it: $P(s,t \mid a,b) = |s\cdot a - t\cdot b|^2/8 = (1 - s\cdot t\cdot(a\cdot b))/4$ is the singlet joint distribution rewritten, for outcome signs $s, t = \pm 1$ at clamp orientations a, b. In that form it yields $E = -\cos\theta$, the half-angle probabilities $\cos^2(\theta/2)$, no-signaling, and order-independence — as it must, being the same distribution.

The derivational content lies one level up. Embedding the law in the family $P \propto |s\cdot a - t\cdot b|^p$, within this ansatz and its stated partially entangled extension, the no-signaling condition singles out $p = 2$ from both sides: any $p \neq 2$ — and any admixture $\varepsilon \neq 0$ of a symmetric deformation — produces a superluminal telegraph, the marginal at one end shifting with the remote setting by $\Delta = 0.146$ at $p = 1$, 0.081 at $p = 3$, 0.072 at $\varepsilon = 0.2$ (at $q = 0.85$, $\Delta = \varepsilon\cdot|(1-2q)^3-(1-2q)|$ for the deformation family). Causality pins $p = 2$ within this ansatz — we make no claim of a general derivation of the Born rule, and the same family contains, at $p \to \infty$, the Popescu–Rohrlich box [PR1994] (CHSH $\to$ 4): no-signaling alone selects neither the quantum correlation nor ours. As we later found, this argument is the measurement-rule analogue of Gisin's no-go for nonlinear quantum dynamics [Gisin1989; Gisin1990]: there, nonlinearity of the evolution enables superluminal signaling; here, nonlinearity of the outcome rule in $\cos\theta$ does — the argument's form is Gisin's, its object is ours. Gisin refutes a nonlinear dynamics; we derive a rule. The two-sided uniqueness of $p = 2$ within the chord family is this program's extension of it.

3.3 The conservation law of the postulate

What causality can do, internal consistency cannot. A symmetric-pair consistency battery — no-signaling, order-independence, probabilities confined to [0,1], and repeatability ($f(\pm 1) = 1, 0$) — is passed by an entire ε-family of non-Born measures; consistency of the pair does not fix the quadratic law. A measure over microconfigurations can nonetheless be constructed: a density of microscopic twist axes with deterministic relaxation reproduces the honest $\cos^2(\theta/2)$ at a single end. But the postulate has then moved from the probability rule into the measure — it relocates between rule, measure, and branch weights, and within the symmetric pair it never annihilates. Only the external principle of causality (3.2) eliminates alternatives.

We call this relocation the conservation of the postulate — a methodological observation within the tested model family, not a consequence of the Kochen–Specker theorem [KS1967]. (Gleason's theorem [Gleason1957] is the nearest formal relative for measure-uniqueness, with its own dimensional premises; we do not invoke it.) The single-end construction stands in the historical line of Kochen–Specker as a demonstration that an outcome measure exists; we draw no inference from that theorem to our observation. The observation shaped every later phase: any mechanism that seemed to derive the Born weights was audited for where it had hidden them instead.

3.4 From analytical layer to dynamical program

Two analytical results sharpened into simulation targets. First, the sawtooth: a uniform microphase measure with conical outcome basins — the natural first reading of the ribbon — gives the exact triangular law $E = 2\theta/\pi - 1$ — the textbook local-realistic correlation at full amplitude; the dynamical model of Section 6 later reached exactly this form, but with amplitude $\rho < 1$ set by source alignment, the analytical layer supplying the limiting shape and the dynamics its measured deficit. This was recorded as an analytical counterexample in the canonical log, and the arrival of the isotropized dynamics at the same form (Section 6.4) was not anticipated. Second, basin measure scaling as $|\text{chord}|^1$ gives the sawtooth while $|\text{chord}|^2$ gives Born — so the question "where does the second power come from?" acquired a geometric face and is posed, formalized but unsolved, in Section 8.3. The simulation program of Sections 4–7 is the systematic attempt to make an elastic, thermal, topologically framed ribbon produce dynamically what the analytical layer showed it must produce structurally — and the measured ways in which it does not.

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4. Methods

4.1 Pre-registration and controls

Every campaign carries a pre-registration file fixing hypotheses, quantitative expectations, and kill-criteria. The commit record proves three properties at three strengths: no pre-registration was ever edited after entering the record (all eight files: one commit each); every pre-registration entered the record before the analysis it governs (raw-data commits precede analysis commits throughout); and for the final campaign, the seed audit, the pre-registration was committed before the runs themselves. We claim the stronger "registered before runs" as working practice throughout; the commit record proves it directly for the last campaign and proves "registered before analysis, never edited" for all. A representative entry (D0, hypothesis D-H2; translated from the Russian original) reads: "$\mathbb{Z}_2$ parity is exactly conserved on accepted trajectories, and the singular-rejection fraction → 0 as dt → 0; kill: if the rejection fraction does not fall with dt, the parity is an artifact of the filter" (D0_prereg.md). Registering the failure condition in advance is what let a null result count as a result rather than as a disappointment to be explained away.

The analysis order was strict: every stage whose result required a fit, a flip, or a model comparison committed its raw measurements before any analysis touched them, as a `*-raw` commit preceding a `*-analysis` commit; the two infrastructure stages, D0 and D1, produced deterministic validations rather than fitted numbers and are single commits (the commit chain is listed in Section 9.1). Every campaign carried controls: an unbiased preparation sampler (boundary axes drawn uniformly on S^2, sector preparation without bias, so the branch is never chosen in advance and the basin decides; measurement.py), mirror pairs (a, b) ↔ (−a, −b), block convergence of the reported observable, an explicit DEGENERATE class for undetermined basins, and — for the central scaling claim — a pre-registered cross-seed audit of the quoted errors (Section 4.2). A standing kill-criterion treated any CHSH > 2 in a model whose readout is local by construction (Section 2.6) as a protocol error to be audited before interpretation; it fired twice, and both times caught a method artifact rather than physics (Section 7).

This machinery was not incidental to the working arrangement but demanded by it. The program was run under an explicit division of roles between generative-AI tooling — design and pre-registration on one side, implementation, runs, and verification on the other — and the human author, who made all decisions and held veto (Section 9.2 and the Supplementary methodological note). Pre-registration, kill-criteria, and the raw-before-analysis order are controls over both AI roles: they bind the design and the execution to commitments made before the data existed, which is precisely the point at which the shared hope for a positive result would otherwise leak into the pipeline.

4.2 Repeatability and reproducibility

Two distinct properties are worth separating. Repeatability is exact: every stage fixes a PRNGKey seed protocol and commits its raw data, so the same seed returns the same number and any figure can be regenerated from a named commit. Reproducibility — whether a result survives a change of seed — is a separate question, and we measured it rather than assumed it: a dedicated seed audit re-ran the $N = 32$ and $N = 96$ cells under twelve fresh base keys and found the cross-seed scatter consistent with the quoted binomial errors ($r = 0.88$ and 1.11; Section 6.1, seed audit commit a26f76b).

The phase history of the program (phases A–D) and the implementation detail — frame algebra, test coverage by invariant, numerical precision — are given in the Supplementary Information (repository: https://github.com/aoktyabrev/ribbon-at-the-bell-bound).

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5. No-go results

The program's negative results are its most durable ones. We state them in increasing order of strength: a gauge-blindness property of axial observables, its sharpening into a structural theorem about where the topological invariant lives, and the census result showing that topology constrains no outcome by measure alone.

5.1 Gauge blindness of axial observables

Every observable used by a physical measurement in this model is axial: a function of the boundary axis $n = R(q)\hat{e}$, where R is the rotation obtained from the frame quaternion q under the double cover $SU(2) \to SO(3)$. Any such function factors through $SO(3)$ by construction, and the kernel of the covering map is exactly the $\mathbb{Z}_2$ we care about: two frame configurations differing only by the lift sign (q vs $-q$) produce identical axial fields everywhere, hence identical statistics for every axial observable. In phase B/C this appeared as an empirical wall — kinks without topological protection relaxed away, and charged configurations, with energies of order $k\_e \cdot (\pi/2..\pi)^2$, remained thermally unpopulated for $T \ll k\_e$. At that stage one could still hope that a cleverer local readout might see the charge.

5.2 The U(1) theorem: axis ≠ frame

Phase D closed that hope. In the R^4 embedding, the boundary clamp fixes an axis, but an axis determines a frame only up to a residual $U(1)$ of rotations about it — and the $\mathbb{Z}_2$ framing class lives precisely in that residual layer, as the boundary sign of the lift. The construction is explicit: the lift-twin test builds pairs of configurations with identical axial fields n_i at every site ($\max |\Delta n| = 0$) and opposite parity. Every candidate readout we constructed — coorientation of the frame against the slice normal, the sign of the nearest interior w-coordinate, windowed relative lift — passed the locality test (stable at window $k \leq 8$) and, without exception, returned identical statistics on the twins. The blindness was not the price of enforcing locality; every local slice datum we could build was blind. We registered this in advance as a named wall ("blind slice"), and it fired. The theorem-level statement: a $\mathbb{Z}_2$ framing invariant of an embedded curve in R^4 is invisible to any boundary observable that factors through the axis, because $\mathbb{Z}_2 \subset \ker(S^3 \to SO(3))$ — the invariant is stored exactly in the fiber that axial physics quotients out. The framing topology thus supplies the $\mathbb{Z}_2$ label and the 4π homotopy (Section 3) — and this theorem closes the other half: for axial measurement, no physical map from that label to outcomes exists. The label is real; the readout that was meant to express it provably cannot. This tension is, we believe, the sharpest single lesson of the program.

5.3 Sector blindness of the axial signal

The complement of 5.2, measured directly: at matched preparation, the axial correlation amplitude is statistically identical in the even and odd topological sectors ($\Delta A = -0.040 \pm 0.038$ at $N = 32$, $M = 1200$). The correlation carried by axial readout is generic chain correlation, controlled by coupling stiffness (Section 6), not by topology. An earlier apparent sector shift (~ 0.07 at $M = 200$) did not survive the statistics increase and is withdrawn.

5.4 The census result: topology forbids no outcome

Pre-registered hypothesis D-H3a proposed that in the odd sector at $\theta = 0$ the aligned outcome branch has empty support — a geometric superselection that would have supplied the exact zeros the Born rule requires. The census refuted it: all 16 cells of $\{\text{sector}\} \times \{\text{branch}\} \times \{\theta \in \{0, \pi\}\}$ contain non-singular representatives, found both numerically (constrained minimization from ≥ 20 diverse starts per cell) and by an explicit homotopy argument — the residual U(1) layer of 5.2 absorbs the framing constraint, so a 2π twist of the fiber connects nominally forbidden configurations to allowed ones. The kill-criterion fired as registered: there is no topological empty-support prohibition in this geometry (non-empty support established by explicit representatives; measure statements are a separate, dynamical question). Combined with 5.2, the two results close both readings of the phase-D bet: topology cannot be read locally, and it forbids nothing globally. What survives of it is one number — the length-stable amplitude (no resolved decay over the measured range) of Section 6 — and the structural explanation of binarity itself.

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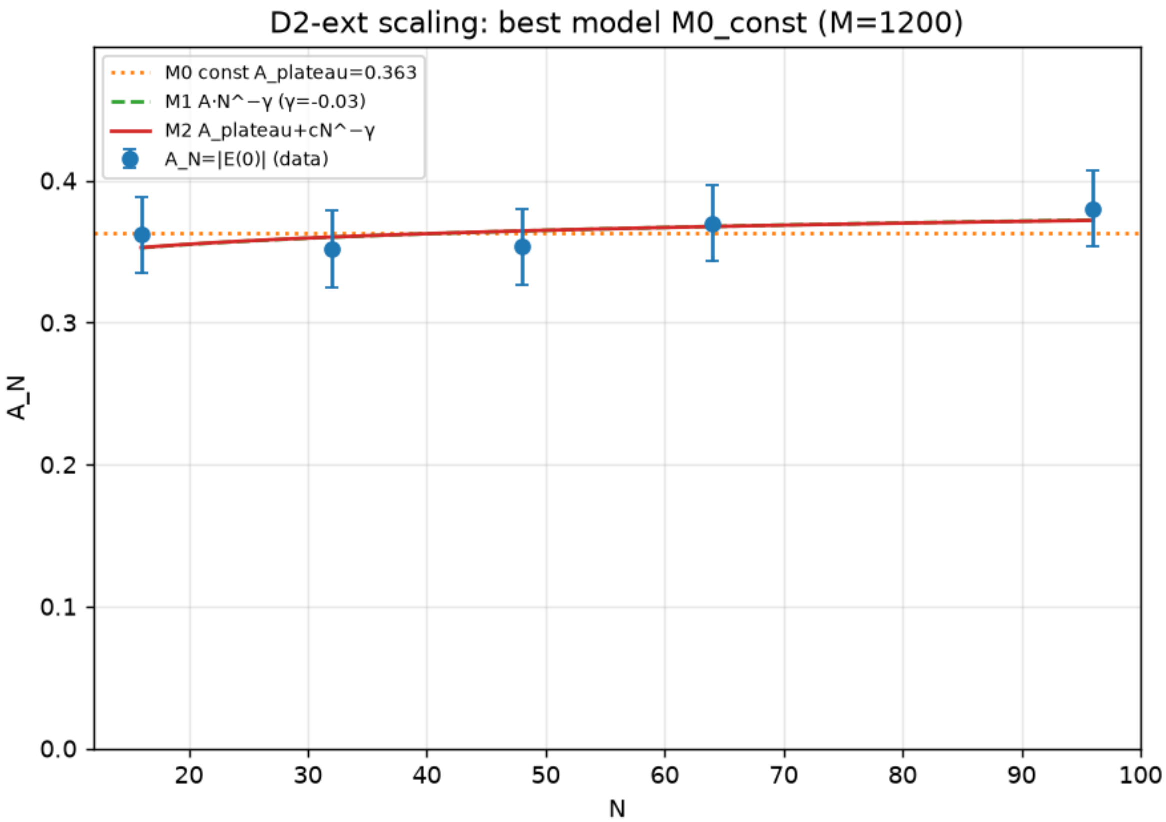
6. Quantitative boundary

6.1 The plateau

The one quantity the ribbon carries that does not dilute with length is the axial readout amplitude of the odd sector. Fit against three models — a constant A_plateau, a power law $A \cdot N^{-\gamma}$, and a saturating $A_{plateau} + c \cdot N^{-\gamma}$ — the constant is substantially preferred among the three tested models ($\Delta AICc = 6.36$): $A_{plateau} = 0.363 \pm 0.012$, with the power law's exponent fitted negative (no decay) (D2-ext;

commit 2784edf). We write A_plateau, and avoid A∞: the limit $N \to \infty$ is not measured; what is measured is no statistically resolved decay over $16 \le N \le 96$. The plateau holds at a second stiffness: cross-scanning $N \in \{16, 32, 64, 96\}$ at k_f×1 and k_f×4, the amplitude change from $N = 16$ to $N = 96$ is $+0.010 \pm 0.038$ and $+0.002 \pm 0.023$ respectively — flat within error at both couplings (DS2; commit f928dd4). The evidence is bounded: $N \in [16, 96]$, k_f ∈ {×1, ×4}. Within that window the plateau is kinetic, not thermodynamic — there is no statistically resolved temperature dependence over $T \in \{0.025, 0.05, 0.10\}$ ($A(T) = 0.368 / 0.397 / 0.353$, within $\sigma \approx 0.027$) (S1-runs; commit a9cef7b) — so it is a property of the basin structure, not of thermal fluctuation. The quoted errors are binomial, and a dedicated seed audit confirms they are honest as a measure of reproducibility: eleven independent repeats of the $N = 32$ cell scatter with $s_{seed} = 0.024$ against a binomial $\sigma = 0.027$, a ratio $r = 0.88$ ($\chi^2 = 7.7 / 10$, $p = 0.66$), and the ratio does not grow with chain length ($r = 1.11$ at $N = 96$) (seed audit; commit a26f76b).



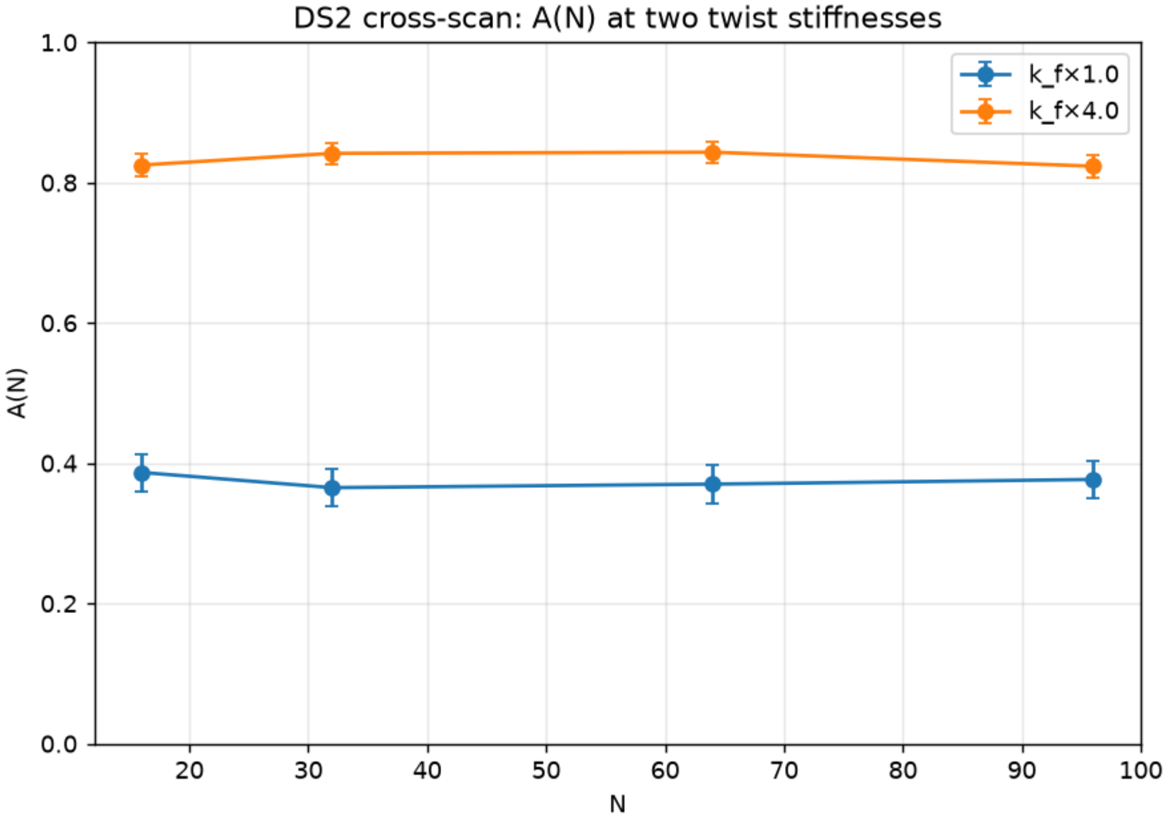


Fig. 1. The plateau. (a) D2-ext: A_N = |E(0)| against chain length N ∈ {16, 32, 48, 64, 96} at M = 1200 replicas, with the three fitted models — constant M0 (A_plateau = 0.363, dotted), power law M1 (γ = −0.03, i.e. fitted growth rather than decay, dashed), and saturating M2 (solid). M0 is substantially preferred by ΔAICc = 6.36 (commit 2784edf); M2 is degenerate on this data and shown for completeness. (b) DS2 cross-scan: A(N) at k_f×1 and k_f×4, flat within error at both stiffnesses (commit f928dd4).

6.2 Origin of the amplitude

The amplitude is stiffness-controlled. Sweeping the twist coupling gives A(k_f) = 0.27 / 0.42 / 0.56 / 0.84 for k_f × {0.5, 1, 2, 4} — a strong monotone dependence that refutes any fixed geometric-measure origin (S1-runs R1; commit a9cef7b). The plateau value A_plateau = 0.363 ± 0.012 belongs to the D2-ext campaign (N ∈ [16, 96] at its baseline setup); the stiffness sweep, run at N = 32, gives 0.418 ± 0.026 at nominal k_f×1 — consistent in scale, not identical — the gap reflects seed-to-seed sampling scatter, quantified in the seed audit (s_seed = 0.024). This only sharpens the point: the amplitude is the reading of a dial, not a constant of the model; the weakness of the signal is a property of soft coupling, not a limit of the mechanism. At θ = 0 the amplitude is, by identity, A = 2·P_aligned − 1 (P_aligned = 0.681 for the D2-ext plateau), so explaining the amplitude means explaining the aligned-basin weight. Two measured facts locate that weight in kinetics rather than thermodynamics. First, it is flat in temperature (Section 6.1), while any Boltzmann reading of the landscape would move with T: the census finds the four branch minima spread over 0.22 in energy (branch means −3.78 for both branches, Δ ≈ 0.006), and no assignment of Boltzmann weights to these energies reproduces a 68/32 split with no resolved temperature dependence (D1 census;

D_synthesis_S1 §2). Second, it is flat in chain length from N = 16 to N = 96 at both measured stiffnesses (Section 6.1), while an equilibrium chain correlation with a finite correlation length would decay. What remains is relaxation itself: the split is set by the geometry of the attraction domains under the clamped dynamics — a domain volume controlled by coupling stiffness. This is the reading fixed in the canonical record after the cross-scan (commit 311d3ae): the amplitude is a stiffness-controlled kinetic correlation, athermal and sector-blind (Section 5.3).

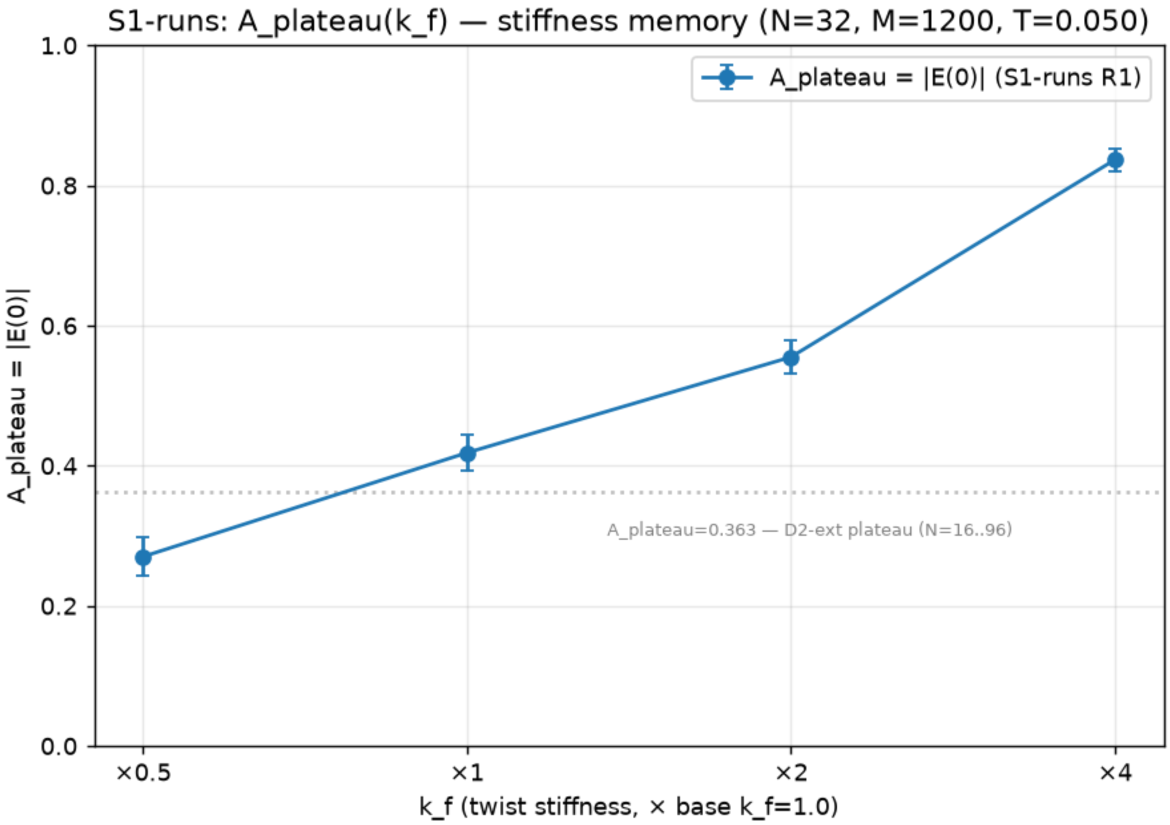


Fig. 2. A_plateau(k_f) — stiffness memory. Four points with 1σ error bars from the frozen S1-runs R1 raw data (N = 32, M = 1200, T = 0.05; commit a9cef7b). Dotted line: the D2-ext plateau A_plateau = 0.363 of Section 6.1, measured over N ∈ [16, 96].

6.3 Anisotropy map

The cosine angular law the correlation appears to follow is anisotropic. Tilting the clamp axis a by α from the privileged axis ê and reading the antiparallel amplitude A(α) = |E_anti|, the signal decays smoothly to zero at α = π/2: from 0.387 to 0.005 at k_f×1 and from 0.865 to 0.008 at k_f×4 (DS3; commit 0fb5452). The best-fit law is cos α at k_f×1 and steepens to ~$\cos^2\alpha$ (an exponential fitting equally well) at k_f×4. The axis ê — simultaneously the twist axis and the readout axis — is strongly privileged; the cosine dependence exists only along it, which is exactly why the isotropic CHSH [CHSH1969] estimator of Section 7 was invalid.

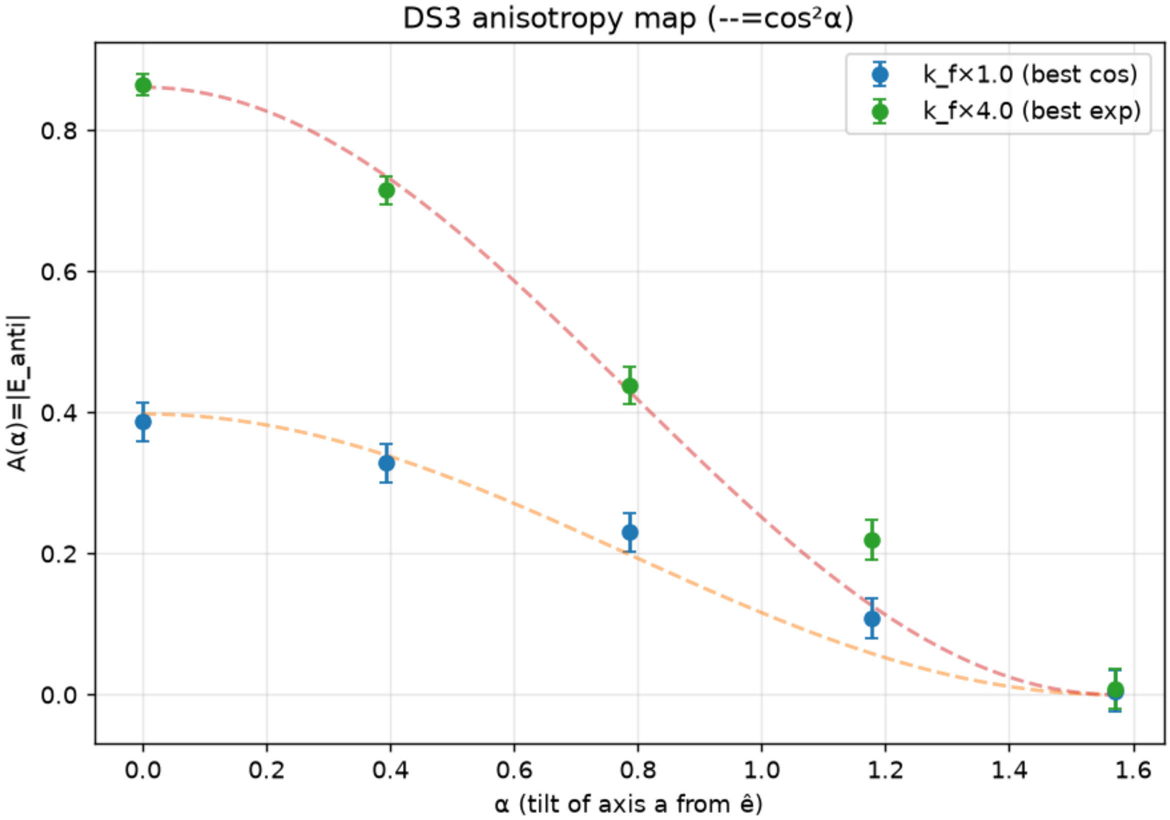


Fig. 3. Anisotropy map. $A(\alpha) = |E\_anti|$ as the clamp axis a is tilted by α from the privileged axis $\hat{e}$, at k_f×1 (best fit cos α) and k_f×4 (best fit exp ≈ $\cos^2\alpha$); dashed curves are $\cos^2\alpha$. The signal decays to zero at $\alpha = \pi/2$ at both stiffnesses: the cosine law exists only along $\hat{e}$ (commit 0fb5452).

6.4 The trilemma and the triangle

Averaged honestly over the shared randomness $\lambda = (R, n\_A, n\_B)$ — where R ~ Haar on SO(3) is the ribbon's orientation, common to both ends through the geometry, while the settings n_A and n_B are drawn independently — the correlation is not a cosine. Fitting the isotropized $E(\theta)$ against both forms, the triangular local-realistic function $E = -\rho(1 - 2\theta/\pi)$ beats the cosine by a wide margin in χ^2: 8 vs 39 at k_f×1 and 1 vs 218 at k_f×4 — the data lie on the straight line, not the curve (DS3; commit 0fb5452). This is the shared-λ local model's signature [Bell1964], and its CHSH value is exactly $S = 2\rho$, with ρ the source-alignment amplitude: $\rho = 0.374 / 0.810$ gives $|S| = 0.75 / 1.62$ at $k\_f \times \{1, 4\}$. The isotropized correlation is statistically consistent with the standard triangular LHV form; we do not claim Bell-locality of the underlying global relaxation dynamics, for which factorization has not been established (Section 2.6(ii)). Three properties therefore trade off and cannot be held together: a cosine form exists only anisotropically (6.3); honest isotropy forces the triangle (this subsection); and the amplitude ρ is bought with stiffness (6.2). The measured family moves toward the Bell bound as stiffness grows: $|S| = 1.62$ at k_f×4 sits a measured distance $2 - 1.62 = 0.38$ short of it. Its deterministic axial limit is expected — not measured — to reach $\rho \to 1$, $|S| \to 2$.

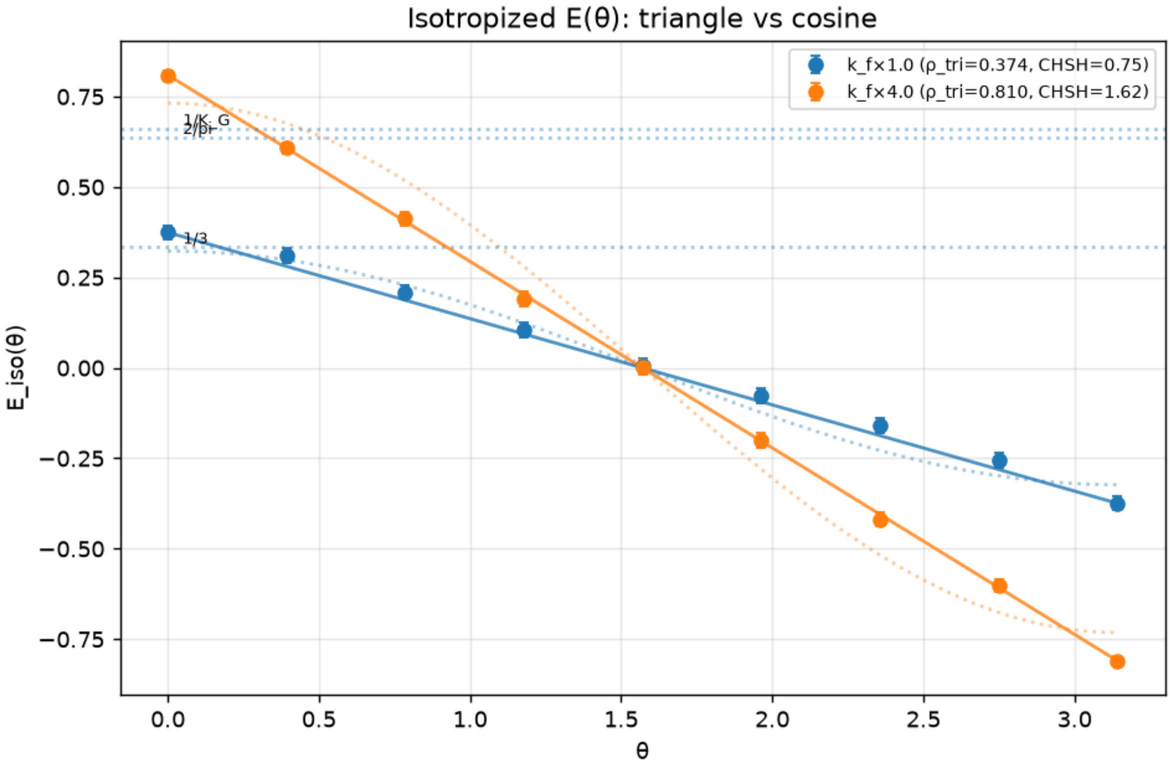


Fig. 4. Triangle versus cosine. The honestly isotropized $E(\theta)$ at k_f×1 and k_f×4: the data lie on the straight triangular law $-\rho(1 - 2\theta/\pi)$ (solid, $\rho = 0.374$ / 0.810, CHSH $= 2\rho = 0.75$ / 1.62), not on the cosine (dotted). Horizontal reference lines mark the literature values $1/3$, $2/\pi$, $1/K\_G$, which were formulated for a cosine LHV and are not direct bounds on a triangular one (DS3 §4; commit 0fb5452).

6.5 Synthesis

The k_f family is a classical frontier: a one-parameter sweep of local models measured from inside the classical region, moving toward the Bell bound as stiffness grows and expected — not measured — to reach CHSH → 2 in its deterministic axial limit. The quantum target — an isotropic cosine with unit amplitude and CHSH $= 2\sqrt{2}$ — lies outside the family on three independent axes at once: amplitude (bounded by $\rho < 1$), form (isotropy forces a triangle, not a cosine), and isotropy itself (the cosine survives only along the privileged axis). No single knob moves the ribbon toward it; that is the quantitative content of the phase-D bet's outcome, and its interpretation is deferred to Section 8.

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7. Case studies in self-correction

A hidden-variable research program faces a specific epistemic hazard: the researcher wants the model to violate a Bell inequality, and the pipeline is built by the same people who hold that hope. We therefore pre-registered, before any CHSH [CHSH1969] campaign, a standing kill-criterion: any result $S > 2$ in a model whose readout is local by construction (Section 2.6) is treated as evidence of a protocol error, triggering a full audit before any interpretation. This criterion fired twice.

Both times it caught an artifact of method, not physics — and both artifacts turn out to be textbook failure modes with decades of history in the Bell-test literature. We report them in detail because we believe the mechanism of their capture is a result in its own right.

7.1 Audit 1: the false-isotropy artifact (DS2)

Applying the standard isotropic CHSH estimator to our stiff-coupling data (k_f×4) yielded |S| = 2.58, comfortably above the classical bound. The pre-registered trigger halted interpretation. Direct measurement of all four correlators — rotating both boundary clamps rather than assuming rotational symmetry — revealed $E(\pi/2, \pi/4) \approx 0.007$ against $E(0, \pi/4) = -0.595$: the system is strongly anisotropic, with the twist axis privileged, and the isotropic estimator is simply invalid for it. The directly measured value is |S| = 1.21, safely sub-classical. The audit had a retroactive cost we accept openly: all previously quoted CHSH values (phase C: 1.48; D2: 0.73–1.25) were computed with the same invalid estimator and are withdrawn. They remain in the canonical record, struck through, with the revision documented (commit 311d3ae).

7.2 Audit 2: the detection loophole, rediscovered from the inside (DS3)

The isotropization protocol — the shared randomness λ = (R, n_A, n_B), with R ~ Haar on SO(3) the ribbon's orientation, common to both ends through the geometry but not available to the experimenter, and the settings n_A, n_B drawn independently (Section 6.4) — produced |S| = 2.39. Since the isotropized protocol is an operational shared-λ scheme with settings independent of λ (Section 2.6(iii)), Bell's theorem [Bell1964] guarantees this is a bug; the trigger fired again. The audit located it in event selection: replicas with a weak axial projection (|proj| < 0.2) were being discarded as DEGENERATE — and the discard rate itself varied with the measurement setting, from 21% of events at the correlation extremes (θ = 0, π) to a peak of 37–38% at intermediate angles (36–37% at the CHSH angles π/4 and 3π/4). This setting-dependence is not incidental to the artifact; it is the artifact. Setting-dependent post-selection is precisely the detection loophole of Pearle [Pearle1970] — the class of local models that forced four decades of loophole-closing in experimental Bell tests. Our pipeline reproduced the disease and, once the cut was removed (all events kept, sign(0) → +1 by convention), the cure: |S| = 1.62 at k_f×4, again sub-classical.

Table 1. Every CHSH value the program produced, and its fate. The isotropic estimator $S = 3E(\pi/4) - E(3\pi/4)$ assumes $E(a, b) = E(|a - b|)$; the ribbon is strongly anisotropic (Section 6.3), so every value computed with it is withdrawn. Withdrawn values are struck through, not deleted (canon "Ревизия CHSH"; commit 311d3ae).

value	campaign	method	status	where corrected
~~≈ 1.48~~	phase C	isotropic estimator	withdrawn: invalid isotropic estimator	superseded by DS2 direct (§7.1)

value	campaign	method	status	where corrected
~~0.73–1.25~~	D2	isotropic estimator	withdrawn: invalid isotropic estimator	superseded by DS2 direct (§7.1)
~~2.39~~	DS3 (primary)	direct, but with setting-dependent DEGENERATE post-selection (|proj| < 0.2, ~36% of replicas)	withdrawn: setting-dependent post-selection — the detection loophole (canon also charges the isotropic estimator: double artifact)	`analysis_ds3`, post-selection removed → 1.62 (§7.2; commit 0fb5452)
1.21	DS2	direct, both ends rotated, no post-selection	valid: direct	—
0.75 / 1.62	DS3	isotropized shared randomness λ = (R, n_A, n_B); CHSH = 2ρ at k_f × {1, 4}	valid: direct	—

No value survives audit above 2; the ribbon is always sub-classical.

7.3 What the two audits establish

First, a negative result of unusual strength: across the entire program, no Bell violation survived an audit. Every S > 2 traced to an identifiable protocol error, and the corrected values sit strictly inside the classical region — converging, in the stiff-coupling limit, toward the Bell bound from below (Section 6). Second, a methodological claim: the pre-registered “too-good-to-be-true” trigger is not decoration. Both artifacts — invalid symmetry assumption, setting-dependent post-selection — are historically documented failure modes that took the community years to identify in experimental contexts. A standing kill-criterion, registered before the data existed and owned by no one at the table, identified both within a single analysis cycle each. Third, a note on provenance: a theoretical entry in our canonical log (the “Bell trap”, §2.7 of the model note) derives that any shared-λ, locally deterministic readout of the ribbon must produce the triangular correlation function — a forced consequence, not a guess; the isotropization campaign landed on exactly that triangle (Section 6). We deliberately rest this point on the logical status of the prediction rather than on its calendar priority, which our commit history cannot independently establish. The program's negative predictions were confirmed by its own later data — which is, we would argue, the behavior one should demand of an honest hidden-variable program before trusting any of its positive claims.

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8. Discussion

8.1 The final lemma

Phase D leaves the program with a lemma sharper than the one it started with. The classical ribbon realizes, from geometry alone: a natural $\mathbb{Z}_2$ structural label with the 4π belt-trick homotopy (the candidate origin of outcome binarity and spinor structure — a candidacy Section 5 shows axial readout cannot redeem), anticorrelation (the singlet sign), a correlation amplitude that does not dilute with chain length (kinetic plateau, two stiffnesses, $N \leq 96$), and a smooth cosine angular law — but only along a privileged axis. It does not realize: exact zeros (no outcome is forbidden by measure, and the invariant that could forbid one is unreadable by axial measurement — Sections 5.4 and 5.2), an isotropic cosine (honest orientation averaging forces the triangular law; Section 6.4), or any Bell violation (every $S > 2$ died in audit; Section 7). The boundary between the two lists is not a fog but a set of named, quantified walls.

8.2 Relation to the classical theorems

The program repeatedly rediscovered, from the inside and with its own numbers, results that mark the classical-quantum boundary. Our causal argument for $\cos\theta$-linearity (the qubit Born rule) is the measurement-rule analogue of Gisin's no-go for nonlinear quantum dynamics [Gisin1989; Gisin1990]: there, nonlinearity of the evolution enables superluminal signaling; here, nonlinearity of the outcome rule in $\cos\theta$ does — the argument's form is Gisin's, its object is ours. The uniqueness of the Born exponent $p = 2$ within the chord family is this program's own extension of the same causal argument; the triangular correlation of the isotropized ribbon is Bell's shared-λ local model [Bell1964], a textbook construction that the program first met as an analytical counterexample (the "saw", session 3) and then found realized dynamically by its own elastic ribbon — the analytical counterexample and the dynamical outcome bracketing the program; the setting-dependent post-selection artifact of DS3 is Pearle's detection loophole [Pearle1970], reproduced and repaired in one audit cycle; and the relocation of the Born postulate between rule, measure, and branch weights — with no elimination available within the symmetric pair itself, only by the external principle of causality (Section 3) — is our own methodological observation within the tested model family, standing in the historical line of Kochen–Specker's demonstration that an outcome measure exists [KS1967], but not derived from that theorem. The number makes the wall tangible twice over: even the family's deterministic limit point ($\rho \to 1$) yields the triangular $E(45°) = -0.500$ — the value the session-3 analytical construction already produced (-0.502, canon §2.7) — against the quantum -0.707; and the measured ribbon does not reach even that, giving -0.187 and -0.405 at the two stiffnesses. We take the convergence itself as evidence: a program that honestly explores a classical ontology does not wander freely — it is funneled onto the same walls the theorems describe, and arrives at them with measured coordinates.

8.3 The open problem

One exit remains formally open, registered in the canonical record as the open problem E: whether a physical fiber-breaking invariant of the R^4 embedding (a second-fundamental-form datum, or a preferred slice normal) can produce an empty aligned-branch basin together with a smooth isotropic cosine — that is, an escape from the trilemma of Section 6.5. We record the obstacles as honestly as the opening — both are our judgment from the phase-D results, not established results themselves: any energetic anchoring of the frame is another stiffness dial, moving the model along the classical family rather than off it; and a shared external frame is shared randomness, bounded by Bell's theorem regardless of its geometric pedigree. A related conjecture from the theory line — that basin measure scaling as the square of the chord ("the ribbon has a face") would reproduce the Born weights — is formalized to the point of an explicit measure, $\mu(B(s,t)) = |s\cdot a - t\cdot b|^2/8$, and posed as an open problem — construct it for all clamp orientations, or prove an impossibility lemma naming the minimal missing structure; it is unsolved, not untested folklore [conjecture]. We consider these open in the technical sense, not the hopeful one.

8.4 The status of the ontology, and of the classical route

The ontological picture itself — the pair as one extended object, whose framing topology supplies the $\mathbb{Z}_2$ label behind binarity and spinority (Section 3.1) — is not refuted by anything in this paper. What is refuted, by construction and at scale, is the proposition that the *studied classical relaxation model* of such an object reproduces singlet statistics. We state our position beyond the minimal one: we regard the classical route as exhausted within the explored model class — framed-curve discretizations with axial readout, over the measured parameter range — and expect further classical modifications inside this class to rediscover the named walls; outside it, Section 8.5 lists what was not tested. The U(1) theorem shows that the topological invariant carrying the ontology's explanatory power is structurally unreadable by axial measurement; the trilemma shows that amplitude, angular form, and isotropy cannot be purchased together anywhere in the accessible family; and the measured family moves toward the Bell bound as stiffness grows ($|S| = 1.62$ at k_f×4), its deterministic axial limit expected — not measured — to reach $S \to 2$. A solution to the open problem E remains the one exit we have not been able to close. If the ribbon picture has a future, it lies in a quantum dynamics of the extended object — a question this paper does not open.

8.5 Limitations

The scope of the evidence is narrower than the scope of the claims a reader might extrapolate, and we state it exactly. The dynamics tested is classical throughout; nothing here bears on a quantum dynamics of the same object. The discretization is one class among the possible: a framed curve in R^4 with axial boundary readout — a triangulated surface, or non-axial boundary couplings, were designed around but not simulated. The U(1) theorem is a statement about axial (SO(3)-factorizable) boundary data, not about all conceivable measurements. And the quantitative results are bounded: $N \in [16, 96]$, k_f $\in \{\times 1, \times 4\}$, one temperature regime verified flat. Within these bounds the walls are measured; beyond them they are extrapolated.

8.6 What we take to be the product

Beyond the specific results, we submit the method as a product in its own right: an ontology driven to its named limit by pre-registration, kill-criteria owned by no one at the table, raw-before-analysis commit discipline, and a standing audit that fires on results that are too good. This machinery caught two textbook artifacts within one analysis cycle each (Section 7) and forced every retreat to be documented rather than forgotten. The program's answer to its opening bet is negative, exact, and reusable — and the same machinery is now free to be pointed at the next idea. Everything should be tested; that is how one either learns something new, or confirms something known — and both are wins.

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9. Code, Data & AI methodology

9.1 Code and data availability

All code, data, and pre-registrations are in the repository: the phase A–C toolkit in `sim/src/ribbon_sim/`, the phase-D model and campaigns in `sim/phase_D/`, and the frozen raw measurements in `sim/phase_D/results/*.json`.

The repository snapshot at submission is archived at Zenodo: DOI 10.5281/zenodo.21388902 (release paper-v1.2). The concept DOI 10.5281/zenodo.21383667 resolves to the latest archived version. Earlier snapshots (paper-v1, pre-camera-ready; paper-v1.1, camera-ready) are retained as prior versions of the same record. To cite the archive:

> Oktiabrev, Artem. *The Ribbon at the Bell Bound: Mapping the Classical Boundary of a Geometric Entanglement Ontology.* Zenodo, 2026. https://doi.org/10.5281/zenodo.21388902

Every stage whose result required a fit, a flip, or a model comparison committed its raw measurements before any analysis touched them, as a `*-raw` commit preceding a `*-analysis` commit: D2 (7a2f18d → 9d7b8e9), D2-ext (624b628 → 2784edf), S1-runs (704e29c → a9cef7b), DS2 (a9f6cfe → f928dd4), DS3 (3d8dd67 → 0fb5452), and the seed audit (2b946ae pre-registration → ecff715 raw → a26f76b analysis). The two infrastructure stages, D0 and D1, produced deterministic validations rather than fitted numbers and are single commits (f25694d, 81cc7da). Each campaign's hypotheses and kill-criteria are fixed in a committed `*_prereg.md`, and what the record proves is graded (Section 4.1): no pre-registration was ever edited after entering it (eight files, one commit each); all entered it before the analysis they govern; and for the seed audit — the one campaign whose pre-registration was committed as its own step (2b946ae) — before the runs as well.

Repeatability is exact: each run script fixes its PRNGKey seed protocol, so the same seed returns the same number, and every figure in this paper can be regenerated from a named commit by running its committed script — `analysis_ext.py` (d2ext_scaling.png), `analysis_ds2.py` (ds2_cross.png), `analysis_ds3.py` (ds3_aniso.png, ds3_iso.png), `plot_s1runs_kf.py` (s1runs_kf.png). Reproducibility — whether a result survives a change of seed — is a separate property, and we measured it rather than assumed it: a pre-registered audit re-ran the $N = 32$ and $N = 96$ cells under twelve fresh base keys, and the cross-seed scatter came out consistent with the quoted binomial errors (r = 0.88 over eleven repeats of the core cell, r = 1.11 at N = 96; seed-audit report, commit a26f76b).

9.2 Use of generative AI

Generative AI tools (Anthropic Claude) were used for research planning and experimental design, software implementation, code review, source checking, and manuscript drafting. The author made all scientific decisions, verified the reported results, and accepts full responsibility for the manuscript.

The detailed working arrangement — the three-party division of roles, the pre-registration and raw-before-analysis order as controls over both AI roles, and the record of errors caught in both directions — is given in the Supplementary Information (repository: https://github.com/aoktyabrev/ribbon-at-the-bell-bound).

9.3 Tooling

No proprietary tooling beyond the AI assistants named was used; all analysis code is in the repository.

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Figure and table list

All figures are produced by committed scripts from committed raw data; each can be regenerated from a named commit (Section 9.1).

#	subject	file	script	§
Fig. 1	plateau A(N)	`d2ext_scaling.png`, `ds2_cross.png`	`analysis_ext.py`, `analysis_ds2.py`	6.1
Fig. 2	A_plateau(k_f), stiffness memory	`s1runs_kf.png`	`plot_s1runs_kf.py`	6.2
Fig. 3	anisotropy map A(α)	`ds3_aniso.png`	`analysis_ds3.py`	6.3
Fig. 4	triangle vs cosine (isotropized)	`ds3_iso.png`	`analysis_ds3.py`	6.4
Table 1	CHSH revision: withdrawn vs valid	inline	—	7.2